

Cover letter

Re: Quotation 20260722-154532811 — Two (2) Counter-Flow Coolers, 8 TPH Soybean Cake, with Fan, Cyclone, Ductwork, Airlock, and Advanced PLC Control Panel

Dear Leigh,

Thank you for the opportunity to quote a cooling system for Pyramid Ag's Memphis, TN facility. Enclosed is our proposal for two (2) counter-flow coolers, each rated for 8 TPH of soybean cake, along with the complete supporting air system — cooling fan, cyclone, ductwork, rotary airlock, and an Advanced PLC control panel for automated operation.

This is a complete, single-source system: Midwest Custom Engineering designed the cooler, fan, cyclone, and control package to work together as one system rather than a collection of separately-sourced components, so performance, airflow balance, and controls integration are our responsibility — not yours to reconcile in the field.

An executive summary of the proposed scope follows this letter. Full technical and commercial detail is included in the attached quotation, reference 20260722-154532811, issued July 22, 2026 and valid through October 20, 2026.

I would welcome the chance to walk through the proposal with you and answer any questions on capacity, control philosophy, or delivery. Please reach out any time at jason@usemce.com.

Sincerely,

Jason Shipley

Midwest Custom Engineering

jason@usemce.com

Executive summary

Soybean Cake Cooling System — Pyramid Ag, Memphis, TN

Project Overview

Pyramid Ag requires cooling capacity for soybean cake processed at its Memphis, TN facility. Midwest Custom Engineering (MCE) proposes two (2) identical counter-flow coolers, each sized for 8 TPH, for a combined 16 TPH of installed cooling capacity. Each cooler is supplied as part of a complete, self-contained air system — fan, cyclone, ductwork, airlock, and PLC controls — engineered to operate as a single integrated unit.

Scope of Supply

Coolers (Qty. 2)	Counter-flow cooler, 8 TPH soybean cake capacity each, engineered for consistent product discharge temperature.
Cooling Fan	Sized to each cooler’s air volume and static pressure requirement for reliable, energy-efficient airflow.
Cyclone	Air/product separator for fines recovery and clean discharge air ahead of exhaust.
Ductwork	Engineered interconnecting ductwork sized and routed for the complete air system.
Airlock	Rotary airlock for continuous, sealed material discharge from the cyclone.
Advanced PLC Control Panel	Automated control of cooler, fan, and airlock operation, with process monitoring and alarm handling.

Key Benefits

- Consistent product quality — counter-flow design and PLC-controlled residence time deliver uniform, controlled discharge temperature for stable storage and shelf life.
- Dust and fines control — the integrated cyclone captures fines from the cooling airstream, protecting product yield and supporting a cleaner operating environment.
- Automated operation — the Advanced PLC panel reduces operator intervention, standardizes startup/shutdown, and provides alarm and diagnostic visibility.
- Single-source responsibility — cooler, fan, cyclone, ductwork, airlock, and controls are engineered together by MCE as one system, eliminating integration risk between separately sourced components.
- Built for continuous duty — equipment is engineered and fabricated for the demands of 24/7 soybean cake processing.

Quotation Summary

Quote Reference	20260722-154532811
Issue Date	July 22, 2026
Expires	October 20, 2026
Currency	USD

Please see the attached formal quotation for complete pricing, technical specifications, and terms. We look forward to supporting Pyramid Ag with this project.



Quote

Quote ref. 20260722-154532811
Issue date Jul 22, 2026
Expires Oct 20, 2026
Currency USD

Seller

Midwest Custom Engineering, Inc.

6526 S Kanner Hwy #215

Stuart, FL 34997

United States

Contact

Jason Shipley (jason@usemce.com)

+16202009109

Buyer

Pyramid Ag

Memphis, TN

Contacts

Leigh Allen (leigh@pyramidag.co)

Effective date	Total discount	Total contract value
On agreement	\$0.00	\$329,502.00

Line items

Name	Quantity	Unit price	Unit discount	Net price
CC 12-289-6	2	\$93,734.00	—	\$187,468.00



- Round Bin Counter Flow Cooler
- 304 Stainless Steel Bin Walls and Top
- 304 Stainless Steel Rotary Feeder With 2HP 3ph/60HZ/480V/Class 2 Div. 2 Motor
- Product Leveler With 2HP 3ph/60HZ/480V/Class 2 Div. 2 Motor
- Three level Indicators for Low Level, Normal Level and High Level
- Air Control Damper
- Mild Steel Full Action Air Operated Louver Discharge Floor
- Mild Steel Hopper Under the Floor
- Door Safety Switch
- Sight Glass
- Explosion Proof Light

Billing summary: Starts on effective date

Fan for Cooler	2	\$16,985.00	—	\$33,970.00
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Fan for cooler

Type: IEPW

Width: 100 %

Speed: 2,839 RPM

Max Design Speed: 2,853 RPM

Volume: 6,400 Acfm

Fan Static Pressure: 20.0 in.wg.

Fan Inlet Air Density: 0.0686 lb/ft³

Inlet Pressure: -20.0 in.wg.

Outlet Velocity: 6,206 fpm

Operating Temp: 90 °F

Elevation: 500 ft ASL

Fan Shaft Power: 33.8 hp

Cold Start Power (68°F): 34.8 hp

Static Efficiency: 48 %

FEI: 0.92

Billing summary: Starts on effective date

HE-36 Cyclone	2	\$13,570.00	—	\$27,140.00
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Name	Quantity	Unit price	Unit discount	Net price
New HE 36 Cyclone, 10 ga, mild steel rated up to 7,600 CFM High Efficiency Centrifugal Collector Model HE-36 • CW inlet rotation • 10 gauge mild steel construction • Exterior welds continuous where possible and cleaned free of weld slag and splatter • 39" Diameter flanged collector housing with riser tube • Flanged 77.25° cone, welded to expansion chamber • Product discharge is flanged 10" round • Flanged rectangular inlet, 25.5" x 10.75" • Draw-thru fitting mounted on riser tube flange, 16.5" x 16.5" exhaust size • Standard mount pads • 8" Dia. bolted inspection port, F/G white sponge gasket • Housing flange gasketing to be Food Grade silicone caulking • Airstream Temperature: 70°F • painted RAL #7012 Basalt Grey • If stand with 40" clearance is desired, please add \$1,950 • If insulated with 1" insulation and aluminum cladding please add \$2,995				

Billing summary: Starts on effective date

PAV 10" Airlock	2	\$10,216.00	—	\$20,432.00
PAV - 10" Airlock AIRLOCK SIZE: 10 HOUSING MATERIAL: CAST IRON END PLATE MATERIAL: CAST IRON BOLT PATTERN: SQUARE SHAFT MATERIAL: MILD STEEL BLADE MATERIAL: MILD STEEL NUMBER OF BLADES: 8 ROTOR CAPACITY: 100% ROTOR FILL EFFICIENCY: 80% VALVE DISPLACEMENT: 0.424CFR ROTOR ENDS: OPEN GASKETS: INCLUDED PACKING: ABRASION RESISTANT PACKING GLANDS: MANUALLY ADJUSTED BEARINGS: SEALED FOR LIFE DRIVE SPECS: CHAIN DRIVE, 0.75HP, 16RPM, 60HZ, 230-460V EXPLOSION-PROOF CLASS 2 DIVISION 2 GROUPS F&G BAFFLE ADAPTER: YES PAINT: WHITE LEAD TIME: 6 WEEKS				

Billing summary: Starts on effective date

Name	Quantity	Unit price	Unit discount	Net price
Est Ductwork	2	\$9,995.00	—	\$19,990.00
Estimated ductwork - final sizing TBD upon site plan layout 16" dia 14 ga Estimate consists of 50' of straight duct and four (4) 90° elbows Billing summary: Starts on effective date				
Control System for Cooler	2	\$20,251.00	—	\$40,502.00
Cooler Control System 460V 3Phase 60Hertz (+/- 10%) (Wye Connection) IDEC Platform Powder Coat Electrical Motor and Control Enclosure - Main Non-Fused Disconnect with Exterior Handle - Safe 24VDC Control Voltage - Safety and Switch Labeling - Wireway Organization - Enclosure Decal - Enclosure Ventilation Certifications, Listings, and Class Rating - cUL Listed under UL508a - Currently a 5 SCCR Rating Maximum (Higher Ratings are Available) - Enclosure will be rated "Type 5" Safety Devices - E-Stop Pushbutton Device - Integrated Safety Relays for Redundant Safety Operation Automation Hardware Assembly 15" Full Color High Resolution Touchscreen - Full Process Depiction with Automated Touch Control - English Units for all Display Values - Continuous Alarm Log and Touchscreen Notifications - Password Based Screen Control Available - Recipe Management and Storage Available if selected - Continuous Data Logging (Sensor Inputs) HMI architecture to be based on a User Tutorial methodology with screen visualizations IDEC PLC for I/O Control - Automated Process Control and Feedback - Powerloss Notification on Restart - Analog and Digital Feedback and Control				

Name	Quantity	Unit price	Unit discount	Net price
5pt Unmanaged Ethernet Switch				
Eaton Spec on VFD				
Cooler Process Automation Functionality				
Automated Cooler Assembly				
System is engineered to safely cool incoming material while maintaining throughput				
Automated Functions				
- Transfer Control				
System is designed to maintain constant throughput of product through the use of multiple actuators				
- Feed and Spreader Control				
Precision motor control creates a consistent material infeed and bed depth				
- Advanced Level Control				
Logic control maintains proper material levels at all times				
- Airflow Control				
System maintains consistent airflow through control dampers				
- Fire Safety				
Control system is designed to safely contain any fire related situations				
- Mechanical System Safeties				
Multiple safety devices are integrated into the system to avoid any mechanical harm				
- Data Feedback				
Advanced control hardware allows for continuous data logging and user feedback in the event of OUT of spec situations				
Inputs per Process				
- Cooler Low Level Indicator (DI)				
- Cooler Mid Level Indicator (DI)				
- Cooler High Level Indicator (DI)				
- Discharge Floor Position Sensor (DI)				
- Exhaust Damper Position Sensor (DI)				
- Exhaust Temp Sensor (AI)				
- Product Discharge Temp Sensor (AI)				
- Cooler Static Pressure (AI)				
- Radar Level Input (AI) (SPARE)				
Safety Inputs per Process				
- Door Switch (DI)				
- Cooler Overload Safety Sensor (DI)				
Outputs per Process				
- Discharge Floor Actuator (HSO)				
- Exhaust Damper Actuator (HSO)				
Automated Motor Control				
- Inlet Rotary Feeder				
3.0HP Motor (VFD Speed Control)				
- Internal material Spreader				
3.0HP Motor (VFD Speed Control)				
- Exhaust Fan				
Communication capability included (VFD Sold Separately)				
- Discharge Floor Linear Actuator				

Name	Quantity	Unit price	Unit discount	Net price
Linear Actuator Control Module and Cables are customer supplied				
3/4HP Motor (Contact Control)				
- Exhaust Air Damper Linear Actuator				
Linear Actuator Control Module and Cables are customer supplied				
3/4HP Motor (Contact Control)				
Billing summary: Starts on effective date				

Subtotal	\$329,502.00
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Discount included in the subtotal	\$0.00
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Due on effective date	\$329,502.00
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The total amount due on the effective date, not including any future payments due later.

Total contract value	\$329,502.00
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Terms

Payments

Customer must agree to the following payment terms: due on receipt

Acceptance

This section is designated for the authorized representatives from both parties involved in this agreement. By signing below, the signatories confirm their acceptance of the terms and conditions outlined in this document.

On behalf of the buyer,

[sig|req|signer1]

(text|req|signer1|letters_only|name)

(text|req|signer1|email_address|email)